Meiosis and Sources of Genetic Variation



What is Variation?

- Differences between organisms within a species.

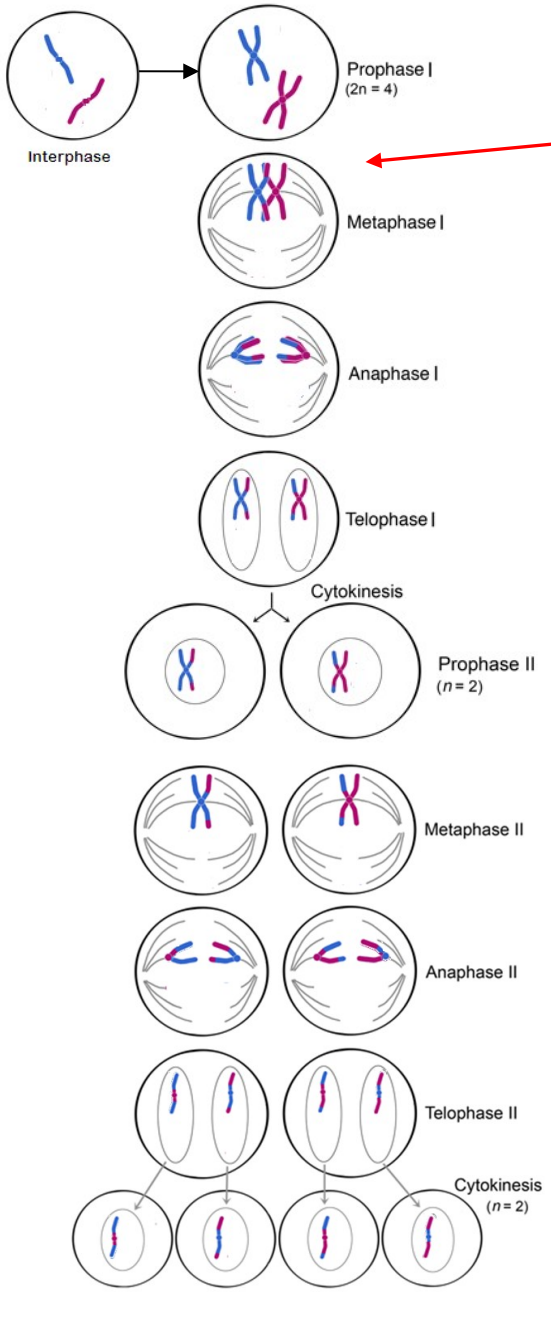
- Human Variation:
  - Height, skin colour, eye colour, weight
  - Combination of genetics and environment

Learning Aim: Define variation

Genetic Sources of Variation

- Random Assortment of Chromosomes during Meiosis
- Crossing Over and Recombination during Meiosis
- Non-Disjunction during Meiosis
- Random Fertilisation
- Mutations

Learning Aim: List genetic sources of variation



Crossing Over!

Meiosis: Showing what happens with a **single homologous pair** of chromosomes.

Remember: this is happening with each of the 23 pairs in each cell that divides.

First Meiotic Division:

- Homologous pairs engage in "crossing over": they exchange a little of each other's DNA.
- Homologous pairs separate, so one of each pair moves to each pole
- At beginning of Prophase II, a haploid karyotype of single chromosomes is present

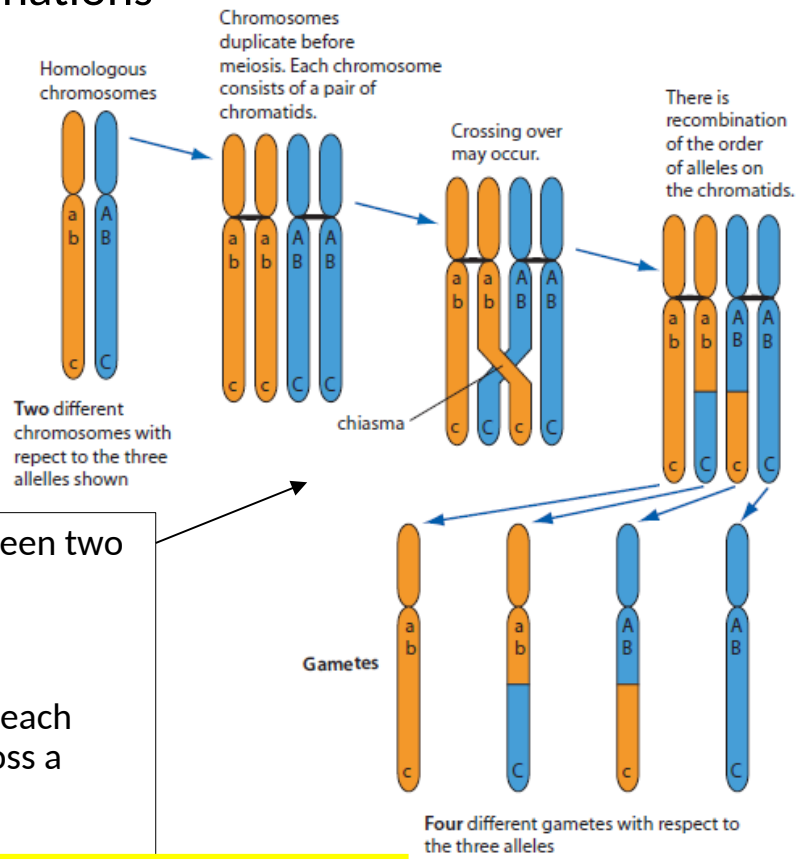
Second Meiotic Division:

- Chromosomes line up on spindle
- Pulled apart during anaphase
- End result is 4 cells, each with a unique haploid karyotype of unduplicated chromatids.

Learning Aim: Describe the main events in the first and second meiotic division

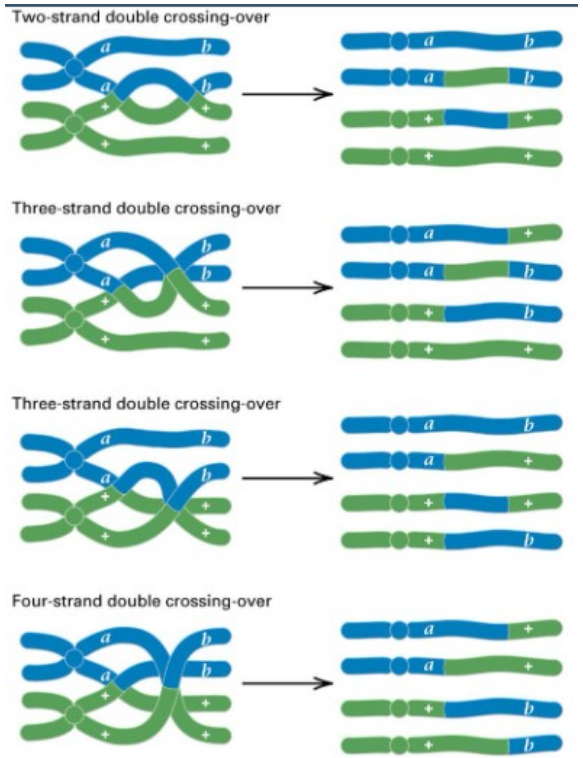
Crossing Over during Meiosis

- Occurs during Meiosis I.
- Homologous pairs exchange some alleles.
- This results in far more gamete combinations than if crossing over didn't occur.



Without crossing over, there would only have been two possible gametes for these alleles.

Crossing over occurs at multiple loci (places) in each chromosome pair. That is a lot of variation across a single karyotype!

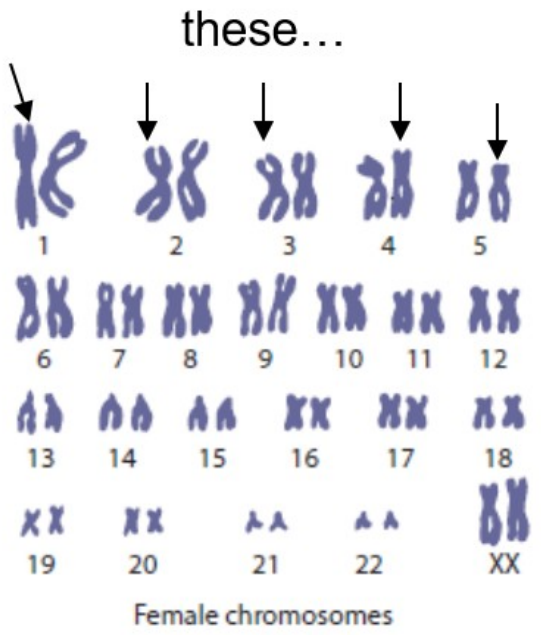


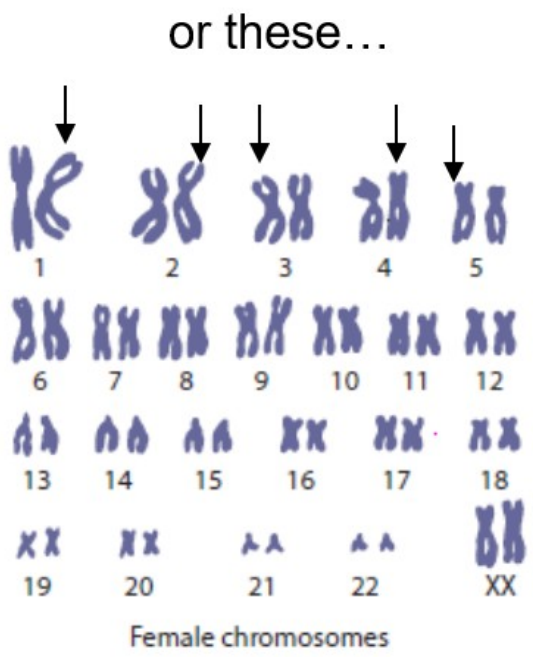
Note: Crossing Over can happen in multiple ways. It is possible to produce a recombinant chromatid in each gamete for a given chromosome.

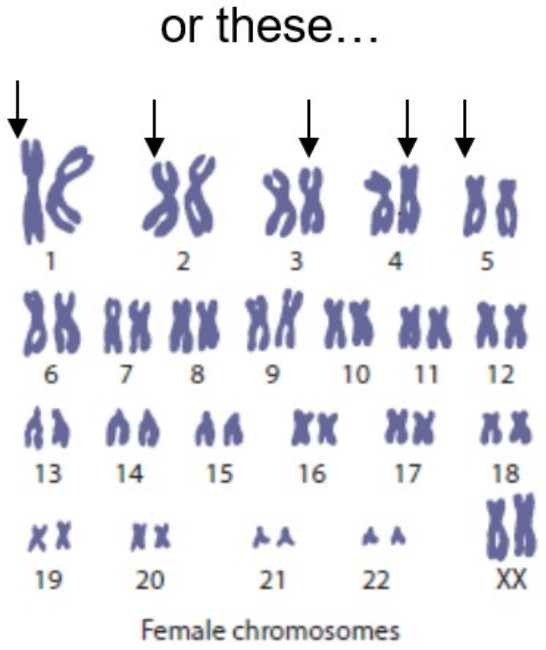
Learning Aim: Describe the process of Crossing Over during Meiosis I

Random Assortment of Chromosomes during Meiosis

- Happens during Meiosis I:
  - Homologous pairs separate: one ends up in each cell at the end of Meiosis I.
  - Karyotype contains 23 chromosome pairs.
  - Lots of possible different combinations depending on which of each pair ends up in a cell, eg:







So, in total, each cell could produce any two of 2^{23} possible random assortments of chromosomes (That is approx 8.5 million combinations)

AND each karyotype is unique also, AND the other gamete that fuses during fertilisation is ALSO one of 8.5 million combinations! That is a LOT of possible offspring karyotypes!

Learning Aim: Describe random assortment of chromosomes during Meiosis

Non-Disjunction During Meiosis

- Where some chromosome pairs do not separate properly during meiosis.
- Can occur during Meiosis I or Meiosis II.
- Results in a Karyotype with an abnormal chromosome number, after fertilisation.
- Generally results in severe birth defects, or miscarriage.
- Generally not an advantageous source of genetic variation.
- Examples include:
  - Down Syndrome
  - Turner's Syndrome
  - Klinefelter's Syndrome
  - Cri-du-chat Syndrome

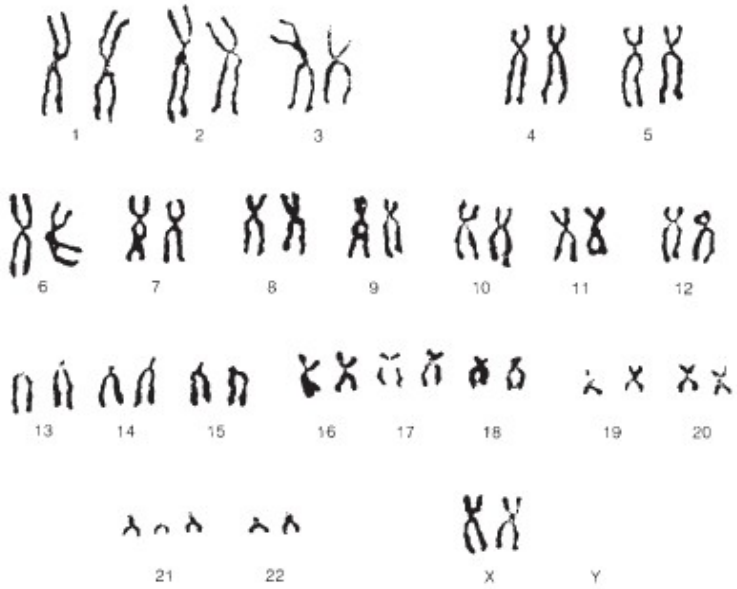


Figure 20.6 Karyotype of Down syndrome (note the extra chromosome 21)



Learning Aim: Describe the effects of non-disjunction during Meiosis

Random Fertilisation

- Refers to the fact that an egg can be fertilised by any one of many different sperm.

- Means more variation in offspring.

Mutations

- Random permanent changes in DNA.
- Variety of causes

Learning Aim: Briefly describe random fertilisation and mutation

New Variations and Survival

- Some variations can be an advantage because:
    - Some of the population is likely to survive during adversity eg:
        - People who store fat easily more likely to survive famine
        - People who have extra body hair more likely to survive an ice age
        - People heterozygous for sickle cell anaemia don’t get malaria as badly.